



Estimating adversarial transferability from feature-importance similarity in explainable intrusion detection systems

Abdlehlah Abdlatef¹ · Marwa Abdulkareem² · Duygu Cakir^{2,3}

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Abstract

Machine Learning (ML)-based Intrusion Detection Systems (IDS) are critical for defending against cyber threats but remain vulnerable to adversarial attacks, posing risks to system reliability and security. This study investigates the transferability of adversarial attacks across XGBoost and LightGBM models in IDS and uses Explainable AI (XAI) techniques to identify features that influence model decisions. Experiments on the CICIDS-2017 and NSL-KDD datasets demonstrate cross-model adversarial transferability, with accuracy reductions of 8.75% and 10.59% for XGBoost and LightGBM, respectively. The XAI-informed analysis further shows that feature-importance patterns can be used to examine model vulnerability under adversarial perturbations. Under direct adversarial testing, XGBoost exhibited a 6.18% reduction in accuracy, illustrating the vulnerability associated with perturbations targeting highly ranked features. Building on prior research linking feature importance to adversarial transferability, this study operationalizes that relationship in the IDS domain through a heuristic formulation based on correlations between model-specific feature-importance vectors. The proposed formulation is intended as a practical approximation for assessing cross-model vulnerability rather than as a deterministic predictor.

Keywords Adversarial attacks · Intrusion detection systems (IDS) · Explainable AI (XAI) · Adversarial transferability · Feature importance

1 Introduction

Intrusion Detection Systems (IDS) have become essential components in cybersecurity frameworks, with machine learning (ML) approaches increasingly adopted to enhance detection capabilities. Despite their performance, ML-based IDS remain vulnerable to adversarial attacks that manipulate input data to evade detection mechanisms [1]. These perturbations to malicious traffic can significantly degrade IDS

accuracy, raising concerns about system reliability in critical applications.

Network IDS attacks are commonly classified as white-box attacks, where attackers have complete knowledge of the target model, or black-box attacks, where attackers have limited information [2]. While white-box attacks are effective but unrealistic in practical scenarios, black-box attacks often achieve lower success rates due to limited transferability between different model architectures. Understanding adversarial example transferability is essential for developing robust ML-based IDS solutions.

Adversarial Machine Learning (AML) examines how machine learning systems respond when targeted by malicious actors attempting to degrade performance or extract sensitive information [3]. AML attacks include data poisoning (corrupting training data), evasion attacks (manipulating input data), and inference attacks (extracting unauthorized information). Common generation techniques include the Fast Gradient Sign Method (FGSM) [4] and Jacobian-based Saliency Map Attack (JSMA) [5] Fig. 1.

To address these threats, researchers have developed defensive mechanisms such as data sanitization and robust

✉ Duygu Cakir
duygu.cakir@bau.edu.tr

Abdlehlah Abdlatef
abdlehlah.abdlatef@bahcesehir.edu.tr

Marwa Abdulkareem
marwaissam.abdulkareem@bau.edu.tr

¹ Graduate School, Bahcesehir University, Istanbul, Besiktas 34353, Turkey

² Department of Software Engineering, Bahcesehir University, Istanbul, Besiktas 34353, Turkey

³ Programme of Applied Mathematics, Galatasaray University, Istanbul, Besiktas 34349, Turkey

classifier development during training, as well as security assessment systems based on game theory and advanced cryptographic techniques during testing [6]. Additionally, Explainable AI (XAI) has emerged as a critical approach for enhancing the interpretability of ML models, particularly in security contexts where understanding model reasoning is crucial [7]. Model-agnostic techniques such as Local Interpretable Model-Agnostic Explanations (LIME) [8] and SHapley Additive exPlanations (SHAP) [6] provide significant tools for interpreting complex model behaviors.

Previous studies have demonstrated the value of XAI methods for identifying influential features, interpreting intrusion detection decisions, and analyzing model vulnerabilities [9, 10]. However, an important gap remains in operationalizing the relationship between feature importance and adversarial transferability for intrusion detection systems. While prior research has examined feature-importance-aware attack generation and transferable adversarial behavior, comparatively limited attention has been given to using feature-importance similarity as an explicit heuristic for approximating cross-model transferability. This gap is particularly relevant for controlled comparisons between structurally related gradient-boosting models using post-hoc XAI methods. Notably, recent work in this journal has applied XAI to black-box attack generation against deep-learning NIDS [11]; the present study is complementary, addressing the distinct question of cross-model transferability between gradient-boosting detectors.

This study addresses this gap by operationalizing the relationship between feature-importance similarity and adversarial transferability in the context of machine learning-based intrusion detection. Rather than claiming to originate the general connection between important features and transferable adversarial behavior, the study adapts this relationship to a controlled comparison of XGBoost and LightGBM models.

The main contributions of this work are as follows:

- We investigate adversarial transferability between two structurally related gradient-boosting models, XGBoost and LightGBM, using the CICIDS-2017 and NSL-KDD datasets.
- We use SHAP and LIME to identify model-relevant features and examine how similarities in feature-importance patterns relate to cross-model adversarial vulnerability.
- We introduce a heuristic quantitative formulation that approximates transferability as a function of the correlation between model-specific feature-importance vectors.
- We empirically compare the heuristic estimate with the observed transferability rate and discuss the limitations arising from the piecewise-constant structure of tree ensembles.
- We provide an IDS-specific operationalization of the broader feature-importance–transferability relationship,

with the scope deliberately restricted to gradient-boosting tree models.

The remainder of this paper is organized as follows: Sect. 2 reviews related work in intrusion detection systems, machine learning techniques for IDS, adversarial attacks, and XAI applications in cybersecurity. Section 3 details our methodology, including the proposed framework, datasets, preprocessing techniques, adversarial example generation, and explainability methods. Section 4 presents experimental results, comprising XAI feature importance analysis, the proposed formulation for adversarial transferability, and model performance evaluation under attacks, and a contextual analysis with existing approaches. Section 5 discusses the empirical findings, the utility and limitations of the proposed formulation, and its relationship to prior adversarial transferability research. Finally, Sect. 6 concludes the paper with a summary of contributions, limitations, and directions for future research.

2 Related work

The domain of intrusion detection has evolved significantly with the integration of machine learning techniques, adversarial methods, and explainability approaches. This section reviews the relevant literature across four key areas: fundamental intrusion detection systems, machine learning applications within IDS, adversarial attacks targeting these systems, and the role of explainable AI in enhancing security solutions. The section then highlights the open questions and research challenges in the field. In particular, this section traces the lineage of feature-importance-based adversarial transferability, from its origins in the computer-vision domain to its recent applications in network intrusion detection.

2.1 Intrusion detection systems

Intrusion detection systems (IDS) have evolved significantly to address the growing sophistication of cyber threats. These systems monitor and analyze network or system behavior to identify suspicious activities, classified primarily as host-based or network-based depending on their deployment location [12]. Network-based IDS (NIDS) examine traffic at network segments, while host-based IDS (HIDS) analyze indicators such as processes and log files on individual machines [13]. Modern IDS implementations have transitioned from signature-based approaches, which struggle with novel threats, to more advanced anomaly-based techniques that model normal behavior and flag deviations [14].

The integration of Big Data Analytics has transformed how organizations handle the overwhelming volume of net-

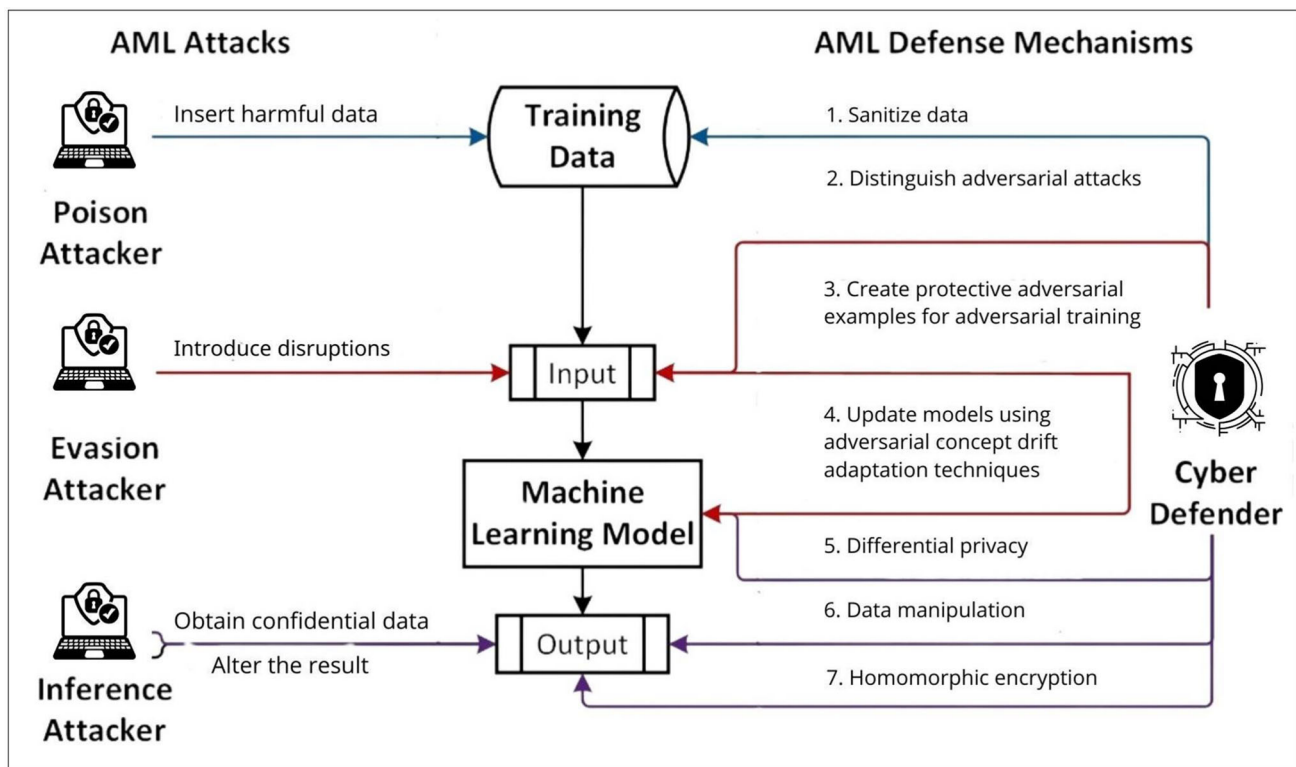


Fig. 1 Adversarial Machine Learning (AML) attack and defense framework. The diagram illustrates the interaction between attackers employing techniques such as FGSM and JSMA to generate adver-

sarial examples, and the defensive mechanisms including adversarial training and input transformation that protect ML-based systems

work traffic data, enabling comprehensive analysis rather than selective logging [15]. This transformation has created both challenges and opportunities for developing more effective intrusion detection mechanisms capable of processing vast data streams while maintaining detection accuracy. Figure 2 shows the steps of an intrusion detection process.

2.2 ML techniques in IDS

Machine learning approaches have been widely adopted in IDS to improve detection capabilities. Traditional ML algorithms such as Support Vector Machines (SVM), Random Forest (RF), and K-Nearest Neighbors (KNN) have demonstrated effectiveness in various intrusion detection scenarios [16]. These approaches typically rely on feature engineering to identify relevant patterns in network traffic data.

Decision tree-based ensemble methods, particularly LightGBM and XGBoost, have shown high performance in intrusion detection tasks due to their ability to handle high-dimensional data with complex relationships. LightGBM's leaf-wise tree growth strategy offers computational advantages over XGBoost's level-wise approach, making it particularly suitable for large-scale IDS applications [17]. While traditional models such as Support Vector Machines

(SVM) continue to be a significant and widely adopted approach in the literature, gradient-boosting ensembles provide shared feature-importance estimation mechanisms that facilitate a controlled study of adversarial transferability.

Deep learning techniques have also been applied to intrusion detection, with architectures such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and hybrid models demonstrating promising results. For instance, Lo et al. [18] developed a CNN-LSTM hybrid model that achieved nearly 100% detection accuracy with minimal false alarm rates for various in-vehicle network attacks. Similarly, Maheswari et al. [19] proposed an optimal cluster-based IDS that achieved 95.01% accuracy on the KDD Cup'99 dataset by combining Multi-segment with Chaotic Manta-ray Foraging Optimization.

2.3 Adversarial attacks on IDS

Adversarial attacks on intrusion detection systems exploit vulnerabilities in machine learning models to evade detection. These attacks can be categorized as white-box attacks, where attackers have complete knowledge of the target system, or black-box attacks, where attackers have limited information [1]. Research by Roshan et al. [20] demonstrated

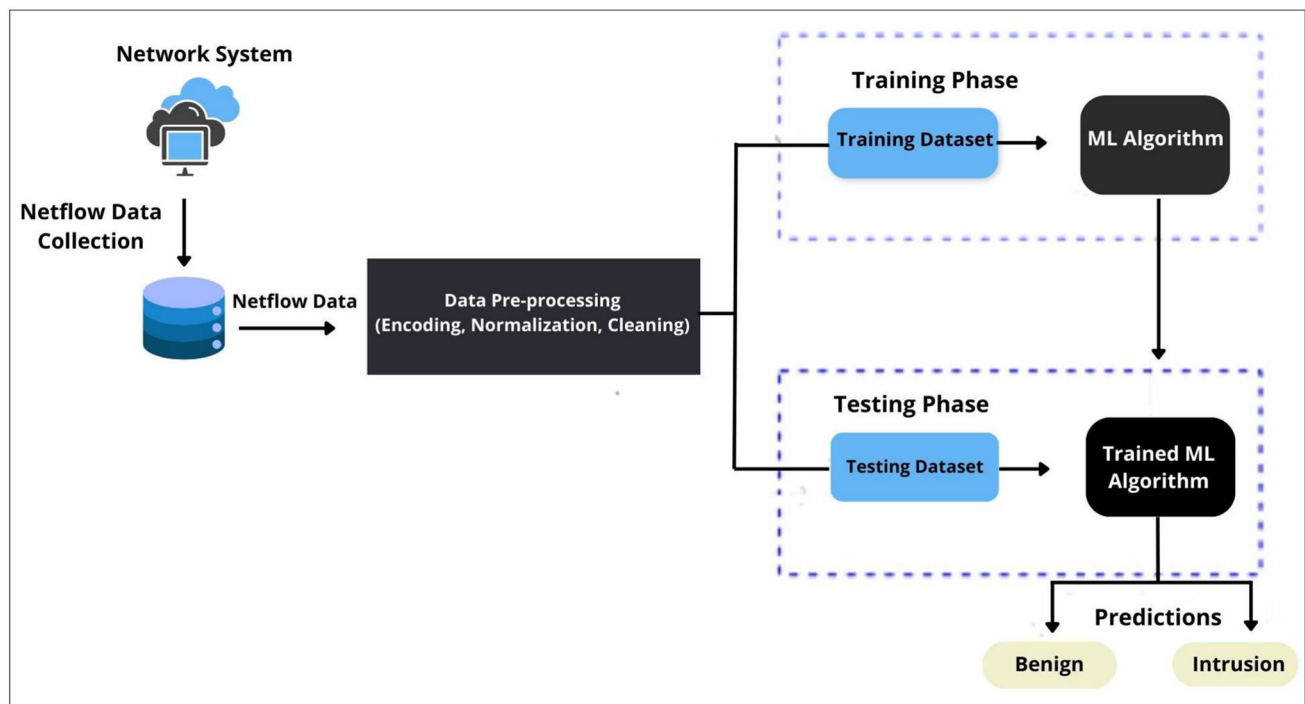


Fig. 2 Steps of an intrusion detection process. The workflow demonstrates the pipeline from network traffic collection through preprocessing, feature extraction, model training, and attack classification

that even small perturbations to network traffic features can significantly degrade the performance of deep learning-based NIDS.

The transferability of adversarial attacks presents a particular challenge, as attacks designed for one model can often successfully evade different models [9]. This phenomenon was observed by Paya et al. [21], who found that adversarial samples generated using Zeroth-order optimization (ZOO) and HopSkipJump attacks could transfer across different classifier architectures with varying success rates.

Generative Adversarial Networks (GANs) have been utilized both to generate adversarial samples and to enhance IDS robustness. Yang et al. [22] introduced DA-GAN, a system that improved IDS performance by creating modified network attack data using GANs and Domain Adaptation, increasing detection rates from approximately 60% to over 99% in various scenarios. While these studies have advanced the understanding of adversarial robustness, they have primarily focused on detection performance without addressing the interpretability of such attacks.

Recent research emphasizes that Explainable AI (XAI) techniques can play a dual role in adversarial contexts [23, 24]. On one hand, explainability methods such as SHAP and LIME can inadvertently expose critical features most influential to model predictions, thereby assisting adversaries in crafting more targeted perturbations [25–27]. On the other hand, these same tools can be used defensively to identify and

mitigate model weaknesses by revealing sensitive regions in the feature space [9, 28–30].

2.4 Feature-importance-based adversarial transferability

A distinct line of research links the importance a model assigns to features with the transferability of adversarial examples across models. This subsection traces that lineage, from its conceptual origin in the computer-vision domain to its recent application in network intrusion detection, in order to position the present contribution within it.

Only limited work has explored how feature-importance correlations influence adversarial transferability between models. The conceptual foundation for this direction originates in the computer-vision domain: Wang et al. [31] proposed the Feature Importance-aware Attack (FIA), arguing that feature importance is strongly correlated with the underlying objects of interest and that this correlation remains largely invariant across different models, which makes importance-guided perturbations transfer more effectively. Our work extends this insight to the intrusion-detection domain along two axes. First, the purpose differs: FIA generates adversarial images that disrupt critical features, whereas we use feature-importance similarity to approximate the transferability of adversarial samples between detectors. Second, the importance signal differs: FIA derives importance

from aggregate gradients over feature maps of a source model, requiring internal access, whereas we rely on post-hoc, model-agnostic XAI (SHAP and LIME), which is better suited to the black-box setting typical of network intrusion detection. This research builds on that cross-domain foundation by operationalizing a heuristic connection between explainability-based feature relevance and the transferability of adversarial attacks.

A closely related line of work uses XAI to guide adversarial attack generation against NIDS. Okada et al. [11] proposed an XAI-driven black-box attack in which KernelSHAP identifies the features most influential to a deep-learning detector, and perturbations are then crafted in the problem space to evade detection. Our study differs from this work in three respects. First, the target model class differs: Okada et al. attack deep feedforward detectors, whereas we focus on gradient-boosting tree ensembles. Second, the objective differs: their framework is concerned with generating successful evasions for a single target, whereas our formulation aims to approximate how readily an adversarial example transfers between two models. Third, the explainability mechanism differs: they rely on KernelSHAP alone, whereas we combine SHAP and LIME to characterize feature-importance similarity across models.

The work most closely related to ours is ETA, the Explainable Transfer-based Black-Box Adversarial Attack framework of Zhang et al. [32]. ETA shares our central premise that adversarial transferability arises when models rely on similar important features, and, like the present study, it targets NIDS, uses CIC-IDS2017, and includes tree-based detectors. Despite this overlap, the two studies are oriented toward different ends. First, in terms of use case, ETA is an attack-generation framework: it produces transferable adversarial examples through an Important-Sensitive Feature Selection procedure grounded in cooperative game theory and perturbation interpretation. Our objective is not to generate stronger attacks but to *approximate* how readily an adversarial example transfers between detectors. Second, in terms of formulation, ETA explains transferability qualitatively but does not provide a quantitative approximation, whereas we introduce an explicit heuristic formulation (Eq. 7) that approximates transferability from the rank correlation of feature-importance vectors. Third, in terms of model scope, ETA spans the broader ML space (including neural networks and tree ensembles), whereas we deliberately restrict our analysis to structurally related gradient-boosting models, which share a common split-gain importance mechanism and thus permit a controlled study of how importance similarity governs transfer. The present work can therefore be read as operationalizing, in a narrower and more analytically controlled setting, the feature-sharing insight examined by ETA and earlier feature-importance-aware transferability research.

Across this line of work, feature importance is consistently treated as a factor associated with cross-model transferability, yet existing studies use this insight to *generate* attacks rather than to *estimate* transferability in advance. The present work addresses that gap by operationalizing the feature-importance/transferability relationship as an explicit heuristic formulation for approximating cross-model transferability, with scope restricted to gradient-boosting detectors for a controlled analysis.

2.5 XAI applications in IDS

Explainable Artificial Intelligence (XAI) has emerged as a critical approach for enhancing the interpretability and trustworthiness of IDS. Arisdakessian et al. [33] conducted a comprehensive survey of XAI techniques applied to anomaly-based IDS within IoT networks, concluding that XAI significantly enhances trust and confidence in cyber defense mechanisms.

SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have been widely applied to interpret IDS decisions. Cui et al. [10] discovered that using these XAI techniques on IDS greatly improved clarity, particularly in distinguishing normal activities from attacks like Denial of Service (DoS). Their research revealed that global explanation methods, such as Permutation Importance and SHAP, boosted model performance by identifying the most important features.

Adversarial attacks on XAI explanations themselves represent an emerging research area. Studies have shown that explanation methods can be vulnerable to manipulations that preserve model outputs while significantly altering the explanations provided to users [7]. This vulnerability underscores the importance of developing robust explanation methods that can withstand adversarial attempts to obscure or mislead interpretations of model behavior.

Recent research has also highlighted that XAI methods themselves may be susceptible to *adversarial manipulations* [23, 34]. Attackers can exploit explanation algorithms to produce misleading yet plausible explanations that preserve the original model output while concealing malicious perturbations [7].

While these studies collectively illustrate the value of XAI in understanding and improving IDS performance, they rarely examine the relationship between explanation-based feature importance and the *transferability of adversarial attacks* across different model architectures. Hence, this research integrates SHAP and LIME within a unified adversarial framework, establishing a heuristic connection between feature importance similarity and cross-model adversarial success, thereby linking interpretability directly to robustness in IDS.

2.6 Open issues and research challenges

Despite significant advancements, several challenges persist in developing robust and interpretable intrusion detection systems. High-quality datasets remain essential for validating IDS models, yet public datasets often contain missing values, incomplete features, or redundant information that reduces model efficacy [35]. Additionally, detection methods require retraining with minimal time investment to identify emerging threats.

The computational cost of training deep learning models presents another challenge, particularly for real-time applications where rapid detection is crucial. Feature selection techniques that reduce dimensionality while preserving discriminative information are essential for optimizing performance [36]. Overfitting remains a significant concern with black-box models like Random Forest, SVM, and CNN, which despite achieving excellent detection rates, often lack transparency.

Transparency and interpretability have emerged as critical requirements for modern IDS, as security analysts increasingly base decisions on system recommendations. Developing novel XAI approaches that provide clear explanations of model behavior represents a significant opportunity for improving trust and understanding in intrusion detection systems [37].

3 Methodology

This research employs an approach to investigate adversarial attack transferability across different machine learning models in intrusion detection systems. Our methodology encompasses framework design, data preparation, adversarial example generation, and explainability techniques. The approach is designed to evaluate how adversarial samples created for one model architecture can affect another, while using explainable AI methods to identify vulnerable features and understand attack mechanisms. The following subsections detail each component of our experimental design and analytical framework.

3.1 Proposed framework

The proposed framework integrates LightGBM and XGBoost models with explainable AI techniques to develop a robust intrusion detection system capable of withstanding adversarial attacks. These two algorithms were deliberately selected because both are gradient-boosting ensemble frameworks that build additive decision trees through iterative optimization. They share similar feature-importance estimation mechanisms based on split gain, which enables a controlled analysis of how feature-importance similarity influences

adversarial transferability. While traditional models such as Support Vector Machines (SVM) and Random Forests (RF) are widely and successfully adopted in intrusion detection research, offering high performance across various network contexts, they utilize fundamentally different decision-making structures (e.g., margin maximization in kernel space or bagging). Consequently, evaluating such heterogeneous models is beyond the scope of the current study, as our framework focuses specifically on transferability guided by the interpretable patterns shared between gradient-boosting architectures.

As illustrated in Fig. 3, the system employs a multi-stage approach combining adversarial testing with XAI-informed defense mechanisms. The system uses the CICIDS-2017 and NSL-KDD datasets to train the LightGBM model for detecting normal and malicious network activities. XAI approaches, specifically SHAP and LIME, are utilized to determine significant features influencing prediction results. These feature importance rankings guide the heuristic-based search methodology that creates adversarial samples by modifying vital features to produce inputs which deceive the model without altering semantic content.

The IDS is evaluated by testing its performance against these adversarial samples, tracking changes in precision, accuracy, recall, and F_1 score metrics. The framework also examines adversarial attack transferability by testing XGBoost-derived adversarial samples against LightGBM to identify vulnerabilities stemming from shared decision boundary areas. This approach allows for a comprehensive assessment of model robustness while maintaining interpretability.

3.2 Datasets and preprocessing

The two primary datasets that have been employed in this research are as follows:

- The **CIC-IDS2017** dataset was collected over a 5-day period based on the behavior of 25 users across HTTP, HTTPS, FTP, SSH, and email protocols. This dataset includes various attack types: Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet, and DDoS. The testbed architecture comprises two isolated networks (Victim-Network and Attack-Network), with the Victim-Network containing essential components such as router, firewall, switches, and multiple devices running Windows, Linux, and macOS.
- The **NSL-KDD V2** dataset represents an enhanced version of the original NSL-KDD dataset, offering improved consistency and a new grouping created by combining certain non-normal classes. This dataset contains 41 network traffic identification attributes categorized as basic features (e.g., duration, protocol type), content-based fea-

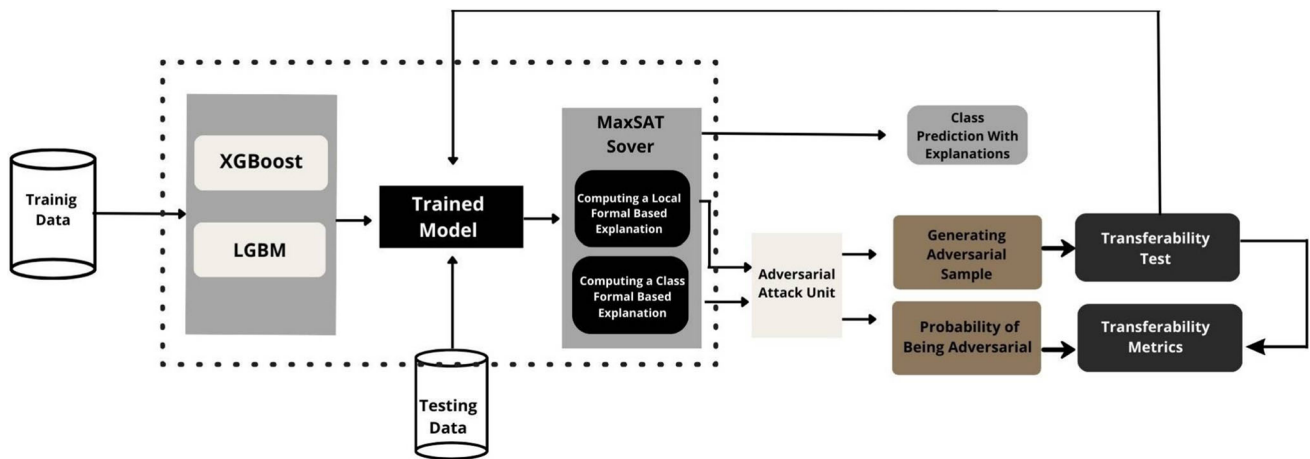


Fig. 3 The proposed framework for explainable adversarial attack detection in intrusion detection systems. The system integrates LightGBM and XGBoost models with SHAP and LIME explanatory techniques to (1) identify critical features influencing detection deci-

sions, (2) generate adversarial samples using feature importance-guided heuristics, (3) evaluate model robustness against these attacks, and (4) assess transferability between different model architectures

tures (e.g., failed logins, root shell access), time-based traffic features (e.g., count, service rate), and host-based traffic features (e.g., destination host count, service rate).

3.2.1 Data preprocessing

Data preprocessing plays a crucial role in ensuring adversarial attack effectiveness. The implemented preprocessing pipeline includes:

1. Data cleaning by removing constant features and quasi-constant features.
2. Applying ordinal encoding to convert non-numeric features into numerical representations.
3. Normalizing numeric features using min-max scaling.
4. Identifying and retaining the most relevant features for model training.

3.3 Adversarial example generation

The research employs a black-box heuristic search approach for generating adversarial samples that target critical features identified through XAI methods. This approach is particularly suitable for real-world scenarios where attackers lack complete knowledge of the target model's architecture and parameters.

3.3.1 Heuristic search method

The heuristic algorithm iteratively modifies input features to find instances that cause model misclassification. Unlike gradient-based methods that require access to model internals, this approach relies on feature importance rankings

from SHAP and LIME to guide the search process. The algorithm systematically perturbs high-importance features within predefined constraints to maximize the probability of misclassification while maintaining the semantic integrity of the original data.

The perturbations are subject to L_p norm constraints to ensure that the modifications remain imperceptible and realistic. Additionally, feature consistency constraints are imposed to preserve the logical relationships between network traffic features, ensuring that the adversarial samples represent plausible network traffic patterns.

3.3.2 Transferability testing

To evaluate the transferability of adversarial attacks, the methodology involves:

1. Identifying critical features driving prediction results in the source model (XGBoost) through SHAP analysis.
2. Creating adversarial samples by manipulating significant features to deceive the source model.
3. Testing these adversarial samples against the target model (LightGBM).
4. Measuring performance degradation using standard metrics (accuracy, precision, recall, F_1 score).
5. Analyzing how the target model processes transferred adversarial samples using interpretability tools.

This approach examines whether shared decision patterns between models enable adversarial samples designed for one system to cause misclassification in another, providing insights into common vulnerabilities across different IDS architectures.

3.4 Explainability methods

The research integrates two primary XAI frameworks to enhance model interpretability:

3.4.1 Local interpretable model-agnostic explanation (LIME)

LIME employs local surrogate models to interpret individual predictions made by the IDS. It operates by perturbing a reference input and evaluating how the model's probability scores change across these perturbed instances. The pixels with the highest weights indicate key attributes influencing the model's decision. LIME is formalized as:

$$\text{Explanation}(x) = \arg \min_{f_s \in F_s} \mathcal{L}(f_0, f_s, \pi_x) + \Omega(f_s) \quad (1)$$

where f_0 is the original model, F_s represents surrogate models, π_x is the proximity measure, and $\Omega(f_s)$ defines surrogate model complexity.

3.4.2 Shapley additive explanations (SHAP)

SHAP quantifies the contribution of each input feature to the output prediction based on Shapley values from coalitional game theory. It fairly allocates the prediction among input features, enabling precise measurement of each feature's importance in model decision-making. The weights for LIFTSHAP $\pi_z()$ are determined by:

$$\pi_z(Z') = \frac{M - 1}{\binom{M}{|Z'|} \times |Z'| \times (M - |Z'|)} \quad (2)$$

where $|Z'|$ represents the number of features in the coalition and M is the maximum coalition size.

3.4.3 Robustness analysis through XAI

The explanations generated by SHAP and LIME serve dual purposes in the framework. First, they identify the features most susceptible to adversarial manipulation, guiding the generation of adversarial samples. Second, they provide insight into how adversarial attacks affect model decision-making by highlighting shifts in feature importance rankings before and after attacks.

4 Results

This section presents the findings from the XAI-based feature-importance analysis, the empirical assessment of the proposed formulation, performance measurements under

adversarial conditions, and the comparison with existing approaches.

4.1 XAI Feature importance analysis

The SHAP and LIME analyses revealed distinct feature importance patterns across both datasets. In the CICIDS2017 dataset, the SHAP summary plot showed that Feature 17 exerted the strongest influence on model predictions, followed by Features 9 and 1, while Features 10 and 5 demonstrated minimal impact. Figure 4 illustrates these global importance rankings.

Local SHAP analysis for individual samples revealed that "Bwd Packet Length Max" and "Init_Win_bytes_backward" were the most influential features directing predictions toward Class 0. The feature "min_seg_size_forward" showed some support for Class 1 but was outweighed by negative features. Similar patterns were observed in LIME explanations, with "Bwd Packet Length Max" (−0.66) and "URG Flag Count" (−0.28) demonstrating substantial influence on directing predictions toward Class 0.

For the NSL-KDD dataset, SHAP analysis identified Feature 36 as the most critical for model predictions, with Features 53 and 21 also showing significant influence, where features 31 and 33 exhibited the least impact (Fig. 5). Local explanations for NSL-KDD samples revealed that "service_remote_job" (0.30) and "service_domain" (0.26) strongly influenced predictions toward Class 1, while "service_hostnames" (−0.69) and "hot" (−0.09) acted as negative influences toward Class 0.

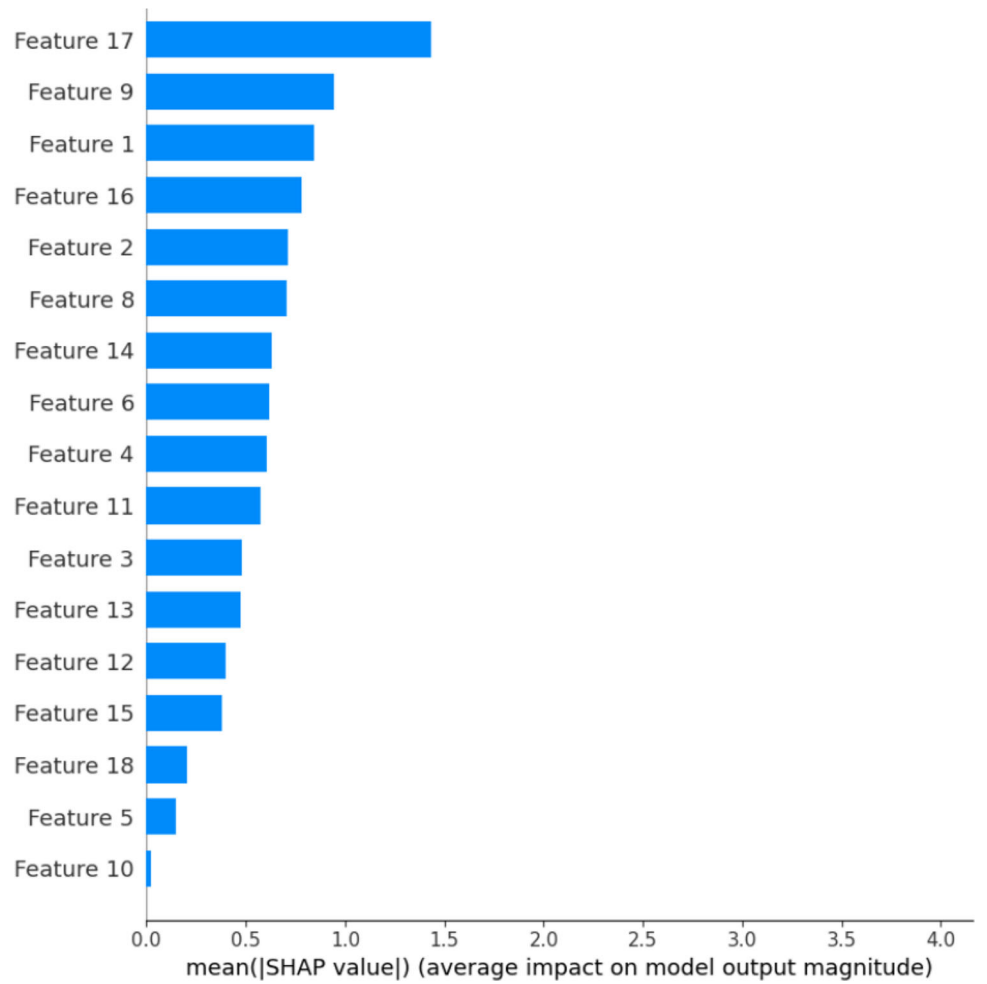
The effectiveness of SHAP and LIME in identifying vulnerable features connects to the theoretical concept of "feature importance sensitivity" described by Ribeiro et al. [38], who theorized that explanation methods can reveal model reliance on specific features that make them susceptible to manipulation. Our findings empirically support this theory, demonstrating that features with the highest SHAP values were indeed the most effective targets for adversarial manipulation, confirming the link between feature importance and adversarial vulnerability.

While the proposed feature-importance analysis highlights the effectiveness of XAI techniques in identifying vulnerable features, an operational formulation is needed to examine how feature-importance similarity may relate to adversarial transferability across IDS architectures. Building on these empirical observations, we introduce a heuristic formulation that approximates this relationship.

4.2 Framework for adversarial transferability

We propose a framework for understanding adversarial transferability in intrusion detection systems. Let f_{θ_A} and f_{θ_B} represent two different machine learning models (e.g., XGBoost

Fig. 4 SHAP summary plot showing global feature importance rankings for the CICIDS2017 dataset. Feature 17 demonstrates the highest impact on model predictions, followed by Features 9 and 1, while Features 10 and 5 show minimal influence



and LightGBM) with parameters θ_A and θ_B , respectively. Both models map the input space $\mathcal{X}$ to the output space $\mathcal{Y}$, where $\mathcal{Y} = \{0, 1\}$ for binary classification in our IDS scenario.

4.2.1 Adversarial example formulation

For a given network traffic sample $\mathbf{x} \in \mathcal{X}$ with true class $y \in \mathcal{Y}$, an adversarial example $\mathbf{x}'$ can be formalized as:

$$\mathbf{x}' = \mathbf{x} + \delta, \text{ such that } f_{\theta}(\mathbf{x}') \neq y \text{ and } \|\delta\|_p \leq \epsilon \quad (3)$$

where δ represents the perturbation vector, $\|\cdot\|_p$ denotes the L_p norm, and ϵ is the maximum perturbation magnitude. In our case, we employ the L_2 norm to constrain perturbations to make sure they remain imperceptible.

4.2.2 Feature importance-guided perturbation

Let $\phi_i(\mathbf{x})$ represent the SHAP value or importance score of feature i for input $\mathbf{x}$. We define a feature importance vector

$\phi(\mathbf{x}) = [\phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots, \phi_d(\mathbf{x})]$ where d is the dimensionality of the input space.

The optimal perturbation vector δ^* can be formulated as the solution to the following optimization problem:

$$\delta^* = \arg \min_{\delta} \|\delta\|_2 \quad \text{subject to} \quad f_{\theta}(\mathbf{x} + \delta) \neq f_{\theta}(\mathbf{x}) \quad (4)$$

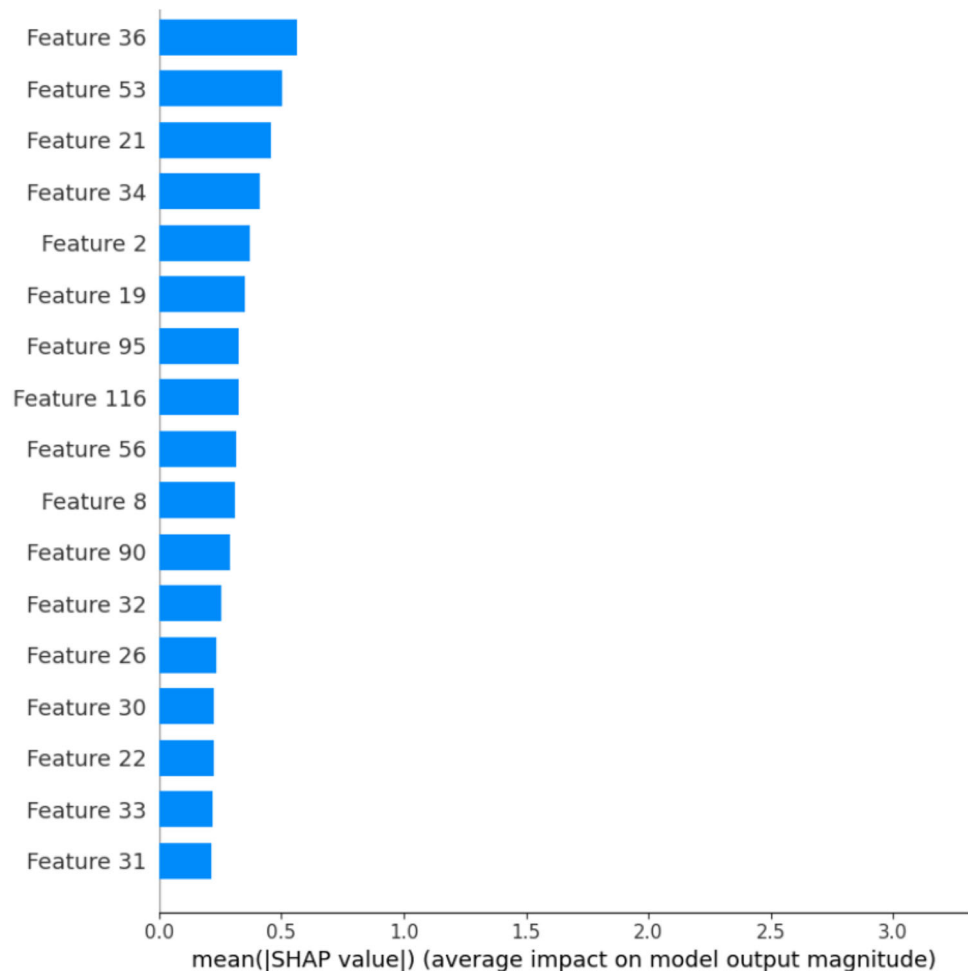
In practice, our heuristic search approach approximates this optimization by prioritizing features with high SHAP values. Specifically, we define a feature mask $\mathbf{m}$ based on feature importance:

$$m_i = \begin{cases} 1, & \text{if } |\phi_i(\mathbf{x})| \geq \tau \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where τ is a threshold parameter. The perturbation is then applied only to the selected features:

$$\delta_i = \begin{cases} \alpha \cdot \text{sign}(\phi_i(\mathbf{x})) \cdot \max(\min(|\phi_i(\mathbf{x})|, \epsilon_i), \epsilon_{\min}), & \text{if } m_i = 1 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

Fig. 5 SHAP summary plot showing global feature importance rankings for the NSL-KDD dataset. Feature 36 as the most critical for model predictions, with Features 53 and 21 also showing significant influence, where features 31 and 33 exhibited the least impact



where α is a scaling factor controlling the strength of perturbation, ϵ_i is the feature-specific constraint, and $\epsilon_{\min}$ is the minimum perturbation magnitude.

4.2.3 Adversarial transferability theorem

We introduce the following theorem to approximate the transferability of adversarial examples between models f_{θ_A} and f_{θ_B} :

Theorem 1 (Adversarial Transferability via Feature Importance *Theorem 1 provides a heuristic foundation for understanding transferability. While tree ensembles like XGBoost and LightGBM produce piecewise constant decision boundaries, they often share similar feature-splitting thresholds when trained on the same data. We suggest that when features show strong correlation in their importance rankings, it may indicate that the models rely on similar feature-ranking patterns and potentially related split structures. With this in mind, Equation (7) should be understood not as a strict geometric transformation, but rather as a probabilis-*

tic approximation that reflects how likely transferability is to occur when models rely on similar features.

Let f_{θ_A} and f_{θ_B} be two different ML models trained on the same task. Let $\rho(\phi_A, \phi_B)$ denote the Spearman rank correlation between the feature importance vectors of models A and B. If an adversarial example $\mathbf{x}'$ is generated using the feature importance vector ϕ_A such that $f_{\theta_A}(\mathbf{x}') \neq f_{\theta_A}(\mathbf{x})$, then the probability of transferability to model B is estimated by the following heuristic approximation:

$$P(f_{\theta_B}(\mathbf{x}') \neq f_{\theta_B}(\mathbf{x})) \approx \frac{1 + \rho(\phi_A, \phi_B)}{2} \cdot P(f_{\theta_A}(\mathbf{x}') \neq f_{\theta_A}(\mathbf{x})) \quad (7)$$

Sketch of Proof The effectiveness of an adversarial example depends on how the perturbation interacts with the decision structure of a model. When two models assign similar importance rankings to the same features, perturbations targeting highly ranked features may affect both models in a similar manner.

As a heuristic assumption, we treat greater similarity between the models' feature-importance rankings as an indicator of greater similarity in their effective feature-space partitions. This assumption is

used only to motivate Equation (7) and does not establish a geometric equivalence between feature-importance correlation and decision-region overlap.

For an adversarial example $\mathbf{x}'$ that changes the prediction of model A, we heuristically approximate the probability that it also changes the prediction of model B as increasing with the similarity of the models' feature-importance rankings. Under this assumption, the scaling factor is approximated by $\frac{1+\rho(\phi_A, \phi_B)}{2}$. $\square$

4.2.4 Empirical assessment

To assess the proposed formulation, we compute the Spearman rank correlation between the feature importance vectors of XGBoost and LightGBM models:

$$\rho(\phi_{\text{XGBoost}}, \phi_{\text{LightGBM}}) = 0.831 \quad (8)$$

Based on the proposed formulation, the estimated transferability rate is:

$$\frac{1 + 0.831}{2} = 0.9155 \quad (9)$$

Our experimental results show a rate of 86.4%. This deviation is expected in a heuristic setting, as Equation (7) captures the probabilistic tendency rather than a deterministic limit. The difference arises from the discrete nature of tree partitions which can cause localized variations in the transferability success that the global correlation ρ does not fully capture.

4.3 Model performance under attacks

Performance metrics for both XGBoost and LightGBM models were evaluated before and after exposure to adversarial attacks. Table 1 presents the results for the CICIDS2017 dataset.

Before adversarial attacks, both models demonstrated excellent performance, with XGBoost achieving 97.41% accuracy and LightGBM reaching 96.22%. However, following the introduction of adversarial samples, performance degraded significantly. XGBoost's accuracy dropped to 91.23% (−6.18 percentage points), while LightGBM's fell to 90.14% (−6.08 percentage points). Similar degradation patterns were observed across precision, recall, and F_1 score metrics.

The adversarial sample generation experiment provided further insights into attack effectiveness. From 10,000 test samples in the NSL-KDD dataset, 3,256 adversarial samples were successfully created (32.56%), with an average perturbation (Euclidean distance) of 1.82. Among the 4,732 attack class samples, 3,165 (66.9%) were misclassified as normal

following perturbation. For the CICIDS-2017 dataset, 3,612 adversarial samples were generated from 12,000 test samples (30.1%), with an average perturbation of 1.67. Among 6,430 attack class samples, 4,112 (63.9%) were misclassified after modification.

Figure 6 demonstrates the vulnerability of both models to adversarial attacks, with a significant portion of malicious samples successfully evading detection after relatively minor feature modifications.

4.4 Transferability of adversarial attacks

Experiments on attack transferability revealed that adversarial samples generated for one model could successfully fool another model architecture. On the CICIDS2017 dataset, performance metrics for both XGBoost and LightGBM decreased substantially when exposed to adversarial samples created for the other model. Table 2 presents these findings.

XGBoost's accuracy decreased by 8.75 percentage points under transferred attacks, while LightGBM exhibited a more significant drop of 10.59 percentage points. Similar degradation patterns were observed for the NSL-KDD dataset, though with slightly less impact: XGBoost's accuracy decreased from 0.9671 to 0.9021 (−6.5 percentage points), and LightGBM's from 0.9663 to 0.8846 (−8.17 percentage points).

The results confirm the existence of shared vulnerabilities between different model architectures, particularly for complex datasets like CICIDS2017. The transferability success indicates that attackers can effectively target multiple IDS implementations using adversarial samples generated for a single model, posing significant security concerns.

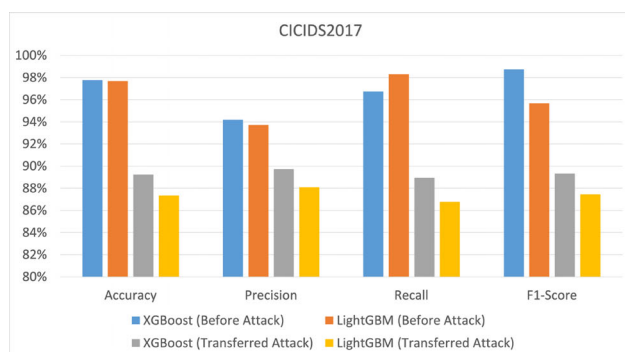
4.5 Comparison with existing approaches

While various studies have investigated adversarial robustness in machine learning-based IDS, direct quantitative comparisons remain challenging due to variations in preprocessing, feature selection, and evaluation protocols. Many studies [2, 6, 39] have demonstrated the vulnerability of models such as XGBoost, Random Forest, and Deep Neural Networks (DNN) to adversarial perturbations on the CICIDS-2017 and NSL-KDD datasets.

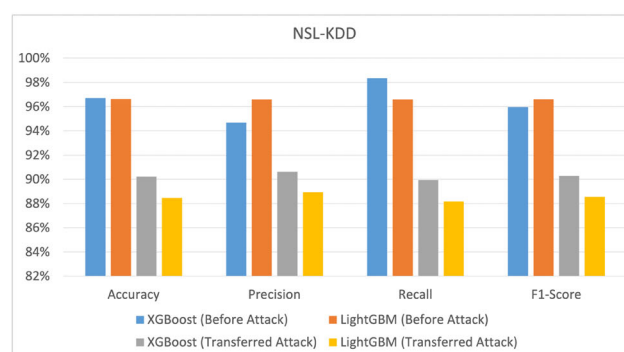
The proposed approach, which integrates a dual-XAI framework using both SHAP and LIME, observed a performance degradation of 6.18%. Rather than claiming absolute superiority, we highlight that this resilience is achieved by leveraging explainability-guided insights during the model evaluation phase. Unlike existing works that primarily utilize a single interpretability method, our dual-XAI approach provides a more comprehensive identification of vulnerable feature regions. Furthermore, the choice of LightGBM facilitates a leaf-wise tree growth strategy that offers specific

Table 1 Performance metrics before and after adversarial attacks on CICIDS2017

Metric	XGBoost Before → After	LightGBM Before → After
Accuracy	0.9741 → 0.9123 (−6.18%)	0.9622 → 0.9014 (−6.08%)
Precision	0.9715 → 0.9107 (−6.08%)	0.9598 → 0.8973 (−6.25%)
Recall	0.9738 → 0.9152 (−5.86%)	0.9635 → 0.9041 (−5.94%)
F_1 score	0.9724 → 0.9129 (−5.95%)	0.9616 → 0.8996 (−6.20%)



(a)



(b)

Fig. 6 Impact of adversarial transferability on model performance metrics. **a** CICIDS2017 dataset results showing performance degradation across accuracy, precision, recall, and F_1 -score when XGBoost and LightGBM models are exposed to adversarial examples generated for the other model. LightGBM exhibits greater vulnerability with a 10.59% accuracy reduction compared to XGBoost's 8.75%. **b** NSL-

KDD dataset results demonstrating similar transferability effects but with less severe degradation, showing the dataset-dependent nature of adversarial robustness. Both charts provide empirical support for the expectation that similar feature-importance patterns may be associated with shared vulnerabilities across the evaluated model architectures

Table 2 Transferability results on CICIDS2017 dataset

Metric	XGBoost Before → Transferred	LightGBM Before → Transferred
Accuracy	0.9778 → 0.8923 (−8.75%)	0.9769 → 0.8735 (−10.59%)
Precision	0.9419 → 0.8973 (−4.73%)	0.9372 → 0.8808 (−6.02%)
Recall	0.9673 → 0.8895 (−8.04%)	0.9830 → 0.8678 (−11.72%)
F_1 score	0.9875 → 0.8934 (−9.53%)	0.9568 → 0.8745 (−8.60%)

computational advantages for high-dimensional network data compared to more query-intensive deep learning architectures.

Traditional gradient-based adversarial attacks, such as the Fast Gradient Sign Method (FGSM) and the Jacobian-based Saliency Map Attack (JSMA), are common benchmarks in the literature. However, these attacks fundamentally rely on direct access to model gradients, making them suitable primarily for white-box scenarios. Our study focuses on explainability-driven transferable attacks where perturbations are guided by feature importance scores instead. This enables a realistic black-box transferability setting, as the attack model does not require access to the target model's internal parameters. Empirical studies by Jmila et al. [40] have shown that gradient-based methods like FGSM often yield limited cross-model transferability compared to strategies guided by feature interpretability.

Recent research by Sharma [41] evaluated query-based black-box attacks, including Zeroth-Order Optimization (ZOO) and HopSkipJump. While these methods can achieve high success rates, they are computationally intensive, often requiring tens of thousands of queries, which limits their applicability to real-time intrusion detection systems. In contrast, the proposed SHAP/LIME-guided framework generates perturbations based on interpretable feature-importance patterns rather than iterative boundary exploration.

5 Discussion

The experimental results demonstrate significant vulnerabilities in machine learning-based intrusion detection systems when subjected to adversarial attacks. The findings align with fundamental frameworks in adversarial machine learning.

According to the linearity hypothesis proposed by Goodfellow et al. [42], machine learning models exhibit vulnerability to adversarial samples because they behave approximately linearly in high-dimensional spaces. The successful generation of adversarial samples with minimal perturbations (average Euclidean distances of 1.82 and 1.67 for NSL-KDD and CICIDS2017, respectively) empirically supports this theoretical prediction.

The performance degradation observed in both XGBoost and LightGBM models corresponds to Szegedy's [43] foundational observation that imperceptible perturbations can cause models to misclassify inputs with high confidence. This phenomenon manifested in the current study with 66.9% and 63.9% of attack samples being incorrectly labeled as benign after adversarial manipulation in the NSL-KDD and CICIDS2017 datasets, respectively. The susceptibility to black-box attacks using heuristic search methods is particularly concerning from a security perspective, as it demonstrates that attackers do not require detailed knowledge of model architecture to craft effective adversarial samples.

The transferability results provide empirical evidence supporting Tramèr's [44] theoretical work on shared adversarial subspaces. According to this theory, different models trained on similar data develop overlapping decision boundaries, creating common vulnerabilities that adversarial samples can exploit regardless of specific model architecture. The observed transferability between XGBoost and LightGBM models—with accuracy reductions of 8.75 and 10.59 percentage points respectively when exposed to transferred attacks—confirms the existence of these shared blind spots, suggesting the possibility of "universal" adversarial samples capable of evading multiple IDS implementations simultaneously.

The effectiveness of feature-importance-guided attacks aligns with the theoretical concept of "feature importance sensitivity" described by Ribeiro et al. [38], who proposed that explanation methods reveal model reliance on specific features that make them susceptible to manipulation. The current findings empirically support this proposal, as features with the highest SHAP values proved to be the most effective targets for adversarial perturbation. This connection between explainability and vulnerability represents a fundamental tension in ML-based security systems: the same features that enable effective classification also create exploitable weaknesses.

The formulation presented in this study builds on previous work by operationalizing a possible relationship between feature-importance similarity and adversarial transferability. The high Spearman correlation (0.831) between the XGBoost and LightGBM feature-importance vectors indicates substantial similarity in their feature rankings. This similarity is consistent with, but does not by itself establish,

the observed cross-model transferability. This formulation provides a probabilistic heuristic for approximating transferability between the structurally similar model pair evaluated in this study. The reasonable agreement between the heuristic estimate (91.55%) and the experimental result (86.4%) supports the utility of the framework for assessing cross-model vulnerability while also indicating that the approximation does not replace direct empirical evaluation. This tension relates to what Papernot et al. [45] described as the "transferability-security" trade-off, where model properties that enhance generalization also facilitate adversarial transferability.

From a theoretical perspective, the integration of XAI methods like SHAP and LIME represents an application of what Miller [46] terms "explanatory debugging"—using explanations to identify and mitigate model weaknesses. The consistency of feature importance rankings across these methods supports their reliability as interpretability tools while revealing potential pathways for defensive strategies. This aligns with theoretical work by Kaur et al. [47], who established that explanation methods can serve as practical tools for model diagnosis and improvement.

While this study focuses on gradient-boosting architectures due to their shared importance estimation mechanisms, the heuristic framework provides a basis for future explorations into how feature relevance informs robustness in heterogeneous models such as SVM or Random Forest.

6 Conclusion

This research has presented an explainability-informed framework for analyzing adversarial vulnerability and approximating cross-model transferability in intrusion detection systems. The findings demonstrate that while machine learning-based IDS can achieve high accuracy under normal conditions, they remain vulnerable to adversarial manipulation, particularly when attackers target features identified as important through explainability methods.

The key contributions of this work include:

- Developing explainable XGBoost and LightGBM-based IDS models that achieve baseline accuracies of 97.41% and 96.22%, respectively, while providing explanations through SHAP and LIME.
- Demonstrating model vulnerability to adversarial manipulation, with successful adversarial samples generated for 32.56% of the NSL-KDD test instances and 30.1% of the CICIDS-2017 test instances.
- Demonstrating adversarial transferability between the evaluated XGBoost and LightGBM models, highlighting shared vulnerabilities within the studied gradient-boosting setting.

- Using SHAP and LIME to identify features associated with adversarial vulnerability and to support the interpretation of model behavior under direct and transferred attacks.
- Operationalizing the relationship between feature-importance similarity and adversarial transferability through a probabilistic heuristic for approximating cross-model vulnerability within the evaluated gradient-boosting model pair.

Despite these advancements, several limitations and opportunities for future work remain. The current approach relies primarily on static datasets, which may not fully capture the dynamic nature of network traffic in operational environments. Additionally, while the proposed defense mechanisms show promise in increasing robustness, they do not completely eliminate vulnerability to adversarial attacks.

Some future research directions we believe that researchers should focus on are as follows:

- Developing advanced data augmentation strategies using generative models like GANs to improve model diversity and robustness.
- Implementing more sophisticated adversarial training techniques that can anticipate and mitigate emerging attack vectors.
- Creating hybrid detection systems that combine deep learning with traditional rule-based approaches to leverage both statistical learning and domain knowledge.
- Optimizing the proposed framework for real-time applications by enhancing scalability and reducing computational costs.
- Further refining XAI methods specifically tailored to adversarial scenarios, improving the understanding of model vulnerabilities.
- While this study focused on gradient-boosting ensembles, future work is needed to assess the extent to which these transferability patterns apply to heterogeneous architectures such as SVM and Random Forest.

Author Contributions M.A. conceptualized the research idea, A.A. implemented the methodology and prepared the figures, and D.C. analyzed the experimental results and wrote the main manuscript text. All authors reviewed the manuscript.

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Declarations

Conflict of interest We declare that there are no conflict of interest related to this manuscript.

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References

1. He, K., Kim, D.D., Asghar, M.R.: Adversarial machine learning for network intrusion detection systems: a comprehensive survey. *IEEE Commun. Surv. Tutor.* **25**(1), 538–566 (2023)
2. Bouaziz, A., Nguyen, M.-D., Valdés, V., Cavalli, A.R., Mallouli, W.: Study on adversarial attacks techniques, learning methods and countermeasures: Application to anomaly detection, pp. 510–517. *ICSOFT* (2023)
3. Chakraborty, A., Alam, M., Dey, V., Chattopadhyay, A., Mukhopadhyay, D.: Adversarial attacks and defences: A survey, (2018). [arXiv:1810.00069](https://arxiv.org/abs/1810.00069) arXiv preprint
4. Lundberg, H.: Increasing the Trustworthiness of AI-based In-Vehicle IDS using Explainable AI, (2022)
5. Hu, Y., Kuang, W., Qin, Z., Li, K., Zhang, J., Gao, Y., Li, W., Li, K.: Artificial intelligence security: threats and countermeasures. *ACM Comp. Surv. (CSUR)* **55**(1), 1–36 (2021)
6. Jiyad, Z.M., Al Maruf, A., Haque, M.M., Gupta, M.S., Ahad, A., Aung, Z.: Ddos attack classification leveraging data balancing and hyperparameter tuning approach using ensemble machine learning with xai. In: 2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T), pp. 569–575 (2024). *IEEE*
7. Sauka, K., Shin, G.-Y., Kim, D.-W., Han, M.-M.: Adversarial robust and explainable network intrusion detection systems based on deep learning. *Appl. Sci.* **12**(13), 6451 (2022)
8. Habib, G., Qureshi, S.: Xai and machine learning for cyber security: A systematic review. *Medical Data Analysis and Processing using Explainable Artificial Intelligence*, pp. 91–104. (2023)
9. Hariharan, S., Rejmol Robinson, R., Prasad, R.R., Thomas, C., Balakrishnan, N.: Xai for intrusion detection system: comparing explanations based on global and local scope. *J. Computer Virol. Hacking Tech.* **19**(2), 217–239 (2023)
10. Cui, J., Zong, L., Xie, J., Tang, M.: A novel multi-module integrated intrusion detection system for high-dimensional imbalanced data. *Appl. Intell.* **53**(1), 272–288 (2023)
11. Okada, S., Mila, H., Akashi, K., Mitsunaga, T., Sekiya, Y., Takase, H., Blanc, G., Nakamura, H.: Xai-driven black-box adversarial attacks on network intrusion detectors. *Int. J. Inf. Secur.* **24**(3), 103 (2025)

12. Abdulganiyu, O.H., Ait Tchakoucht, T., Saheed, Y.K.: A systematic literature review for network intrusion detection system (ids). *Int. J. Inf. Secur.* **22**(5), 1125–1162 (2023)
13. Habeeb, M.S., Babu, T.R.: Network intrusion detection system: a survey on artificial intelligence-based techniques. *Expert. Syst.* **39**(9), 13066 (2022)
14. Afolabi, A.S., Akinola, O.A.: Network intrusion detection using knapsack optimization, mutual information gain, and machine learning. *J. Electr. Computer Eng.* **2024**(1), 7302909 (2024)
15. Thakkar, A., Lohiya, R.: A survey on intrusion detection system: feature selection, model, performance measures, application perspective, challenges, and future research directions. *Artif. Intell. Rev.* **55**(1), 453–563 (2022)
16. Abdulganiyu, O.H., Tchakoucht, T.A., Saheed, Y.K.: Towards an efficient model for network intrusion detection system (ids): systematic literature review. *Wireless Netw.* **30**(1), 453–482 (2024)
17. Mallampati, S.B., Seetha, H., Batchu, R.K.: Pcb-lgbm: A hybrid feature selection by pearson correlation and boruta-lgbm for intrusion detection systems. In: *International Conference on Computational Intelligence and Data Engineering*, pp. 523–533. Springer (2022)
18. Lo, W., Alqahtani, H., Thakur, K., Almadhor, A., Chander, S., Kumar, G.: A hybrid deep learning based intrusion detection system using spatial-temporal representation of in-vehicle network traffic. *Veh. Commun.* **35**, 100471 (2022)
19. Maheswari, K., Siva, C., Priya, G.N.: An optimal cluster based intrusion detection system for defence against attack in web and cloud computing environments. *Wireless Pers. Commun.* **128**(3), 2011–2037 (2023)
20. Roshan, K., Zafar, A., Haque, S.B.U.: Untargeted white-box adversarial attack with heuristic defence methods in real-time deep learning based network intrusion detection system. *Comput. Commun.* **218**, 97–113 (2024)
21. Paya, A., Arroni, S., García-Díaz, V., Gómez, A.: Apollon: a robust defense system against adversarial machine learning attacks in intrusion detection systems. *Computers Secur.* **136**, 103546 (2024)
22. Yang, Y.-G., Fu, H.-M., Gao, S., Zhou, Y.-H., Shi, W.-M.: Intrusion detection: a model based on the improved vision transformer. *Trans. Emerg. Telecommun. Technol.* **33**(9), 4522 (2022)
23. Baniecki, H., Biecek, P.: Adversarial attacks and defenses in explainable artificial intelligence: a survey. *Information Fusion* **107**, 102303 (2024)
24. Noppel, M., Wressnegger, C.: Sok: Explainable machine learning in adversarial environments. In: *2024 IEEE Symposium on Security and Privacy (SP)*, pp. 2441–2459. IEEE (2024)
25. Slack, D., Hilgard, S., Jia, E., Singh, S., Lakkaraju, H.: Fooling lime and shap: Adversarial attacks on post hoc explanation methods. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pp. 180–186. (2020)
26. Aïvodji, U., Hara, S., Marchand, M., Khomh, F.: Fooling shap with stealthily biased sampling. In: *The Eleventh International Conference on Learning Representations*, (2022)
27. Ghorbani, A., Abid, A., Zou, J.: Interpretation of neural networks is fragile. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 3681–3688. (2019)
28. Mahbooba, B., Timilsina, M., Sahal, R., Serrano, M.: Explainable artificial intelligence (xai) to enhance trust management in intrusion detection systems using decision tree model. *Complexity* **2021**(1), 6634811 (2021)
29. Pinhasov, B., Lapid, R., Ohayon, R., Sipper, M., Aperia, Y.: Xai-based detection of adversarial attacks on deepfake detectors, (2024). [arXiv:2403.02955](https://arxiv.org/abs/2403.02955) arXiv preprint
30. Vreš, D., Robnik-Šikonja, M.: Preventing deception with explanation methods using focused sampling. *Data Min. Knowl. Disc.* **38**(5), 3262–3307 (2024)
31. Wang, Z., Guo, H., Zhang, Z., Liu, W., Qin, Z., Ren, K.: Feature importance-aware transferable adversarial attacks. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 7639–7648. (2021)
32. Zhang, H., Han, D., Zhuang, S., Wang, Z., Sun, J., Liu, Y., Liu, J., Dong, J.: Explainable and transferable adversarial attack for ml-based network intrusion detectors. *IEEE Trans. Dependable Secure Comput.* **22**(5), 5090–107 (2025)
33. Arisdakessian, S., Wahab, O.A., Mourad, A., Otrók, H., Guizani, M.: A survey on iot intrusion detection: Federated learning, game theory, social psychology, and explainable ai as future directions. *IEEE Internet Things J.* **10**(5), 4059–4092 (2022)
34. Wang, J., Wu, Y., Li, M., Lin, X., Wu, J., Li, C.: Interpretability is a kind of safety: An interpreter-based ensemble for adversary defense. In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 15–24 (2020)
35. Muneer, S., Farooq, U., Athar, A., Ahsan Raza, M., Ghazal, T.M., Sakib, S.: A critical review of artificial intelligence based approaches in intrusion detection: a comprehensive analysis. *J. Eng.* **2024**(1), 3909173 (2024)
36. Agalit, M.A., Sadiqui, A., Khamlichi, Y., Chakir, E.M.: Hybrid intrusion detection system for wireless networks. In: *WITS 2020: Proceedings of the 6th International Conference on Wireless Technologies, Embedded, and Intelligent Systems*, pp. 507–513. Springer (2021)
37. Altulaihan, E., Almaiah, M.A., Aljughaiman, A.: Anomaly detection ids for detecting dos attacks in iot networks based on machine learning algorithms. *Sensors* **24**(2), 713 (2024)
38. Ribeiro, M.T., Singh, S., Guestrin, C.: “Why should i trust you?” Explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144 (2016)
39. Oseni, A., Moustafa, N., Creech, G., Sohrabi, N., Strelzoff, A., Tari, Z., Linkov, I.: An explainable deep learning framework for resilient intrusion detection in iot-enabled transportation networks. *IEEE Trans. Intell. Transp. Syst.* **24**(1), 1000–1014 (2022)
40. Jmila, H., Khedher, M.I.: Adversarial machine learning for network intrusion detection: a comparative study. *Comput. Netw.* **214**, 109073 (2022)
41. Sharma, S.: Adversarial attacks against network intrusion detection systems, Purdue University (2024). (Master’s thesis)
42. Goodfellow, I.J., Shlens, J., Szegedy, C.: Explaining and harnessing adversarial examples, (2014). [arXiv:1412.6572](https://arxiv.org/abs/1412.6572) arXiv preprint
43. Szegedy, C., Zaremba, W., Sutskever, I., Bruna, J., Erhan, D., Goodfellow, I., Fergus, R.: Intriguing properties of neural networks, (2013). [arXiv:1312.6199](https://arxiv.org/abs/1312.6199) arXiv preprint
44. Tramèr, F., Papernot, N., Goodfellow, I., Boneh, D., McDaniel, P.: The space of transferable adversarial examples. [arXiv preprint arXiv:1704.03453](https://arxiv.org/abs/1704.03453) (2017)
45. Papernot, N., McDaniel, P., Jha, S., Fredrikson, M., Celik, Z.B., Swami, A.: The limitations of deep learning in adversarial settings. In: *2016 IEEE European Symposium on Security and Privacy (EuroS&P)*, pp. 372–387 (2016). IEEE
46. Miller, T.: Explanation in artificial intelligence: insights from the social sciences. *Artif. Intell.* **267**, 1–38 (2019)
47. Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H., Wortman Vaughan, J.: Interpreting interpretability: understanding data scientists’ use of interpretability tools for machine learning. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–14. (2020)